ADMS

16.25.1

Geographic Network View Symbology Reference Guide

Copyright

Change History

Date	Change Description
May 31, 2023	ADMS 3.16: • Editorial and formatting updates.
Oct 20, 2023	ADMS 16.1.0: • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
Mar 25, 2024	ADMS 16.7.0: • Converted document to new format.

Contents

1. Overview

1.1. Default View

This section provides a high-level overview of the electrical devices, halos, and background shapes visible on the geographic network view at the default zoom level (i.e., zoom = 1).

At normal zoom level (whole feeder view), smaller network elements (such as lateral lines, distribution transformers, fuses, fault indicators, and cabinets) are hidden. The voltage violation and overload halos are applied to the entire switchable sections that contain devices that are in violation (see Figure 1). Only large background shapes are visible (see Figure 2).

The visible zoom levels for all network elements are configurable in the Viewer Configuration Editor (for more details, refer to the *Viewer Configuration Editor User Guide*).

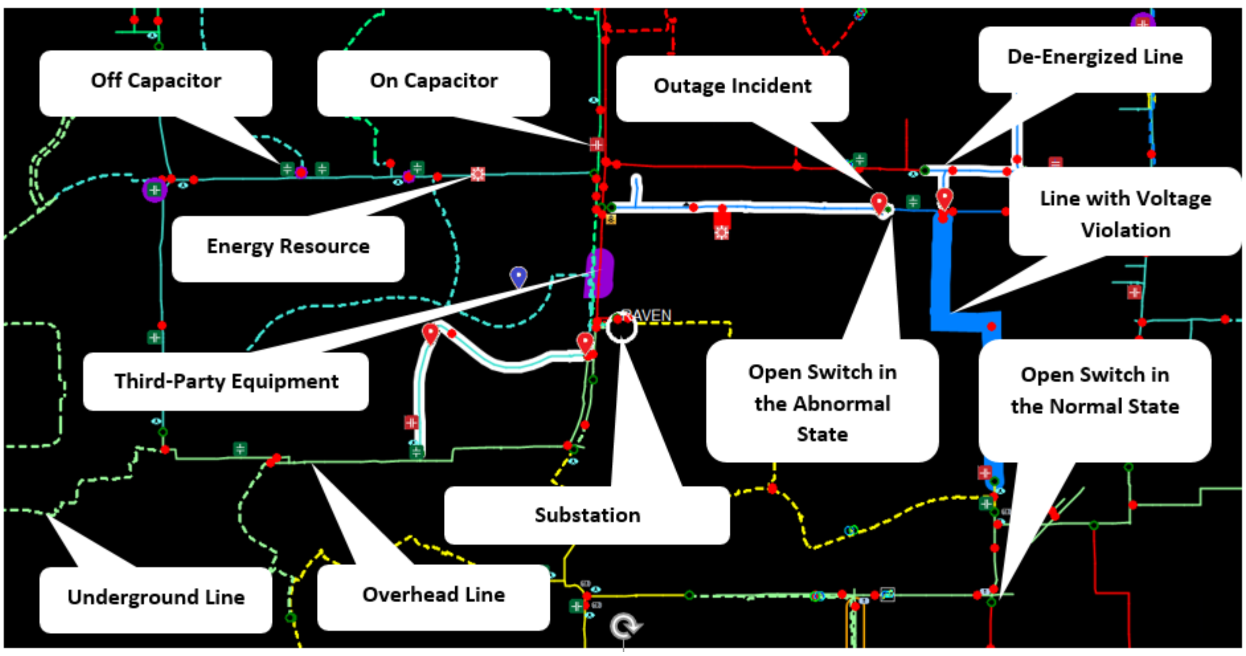


Figure 1. Network Elements at Default Zoom Level

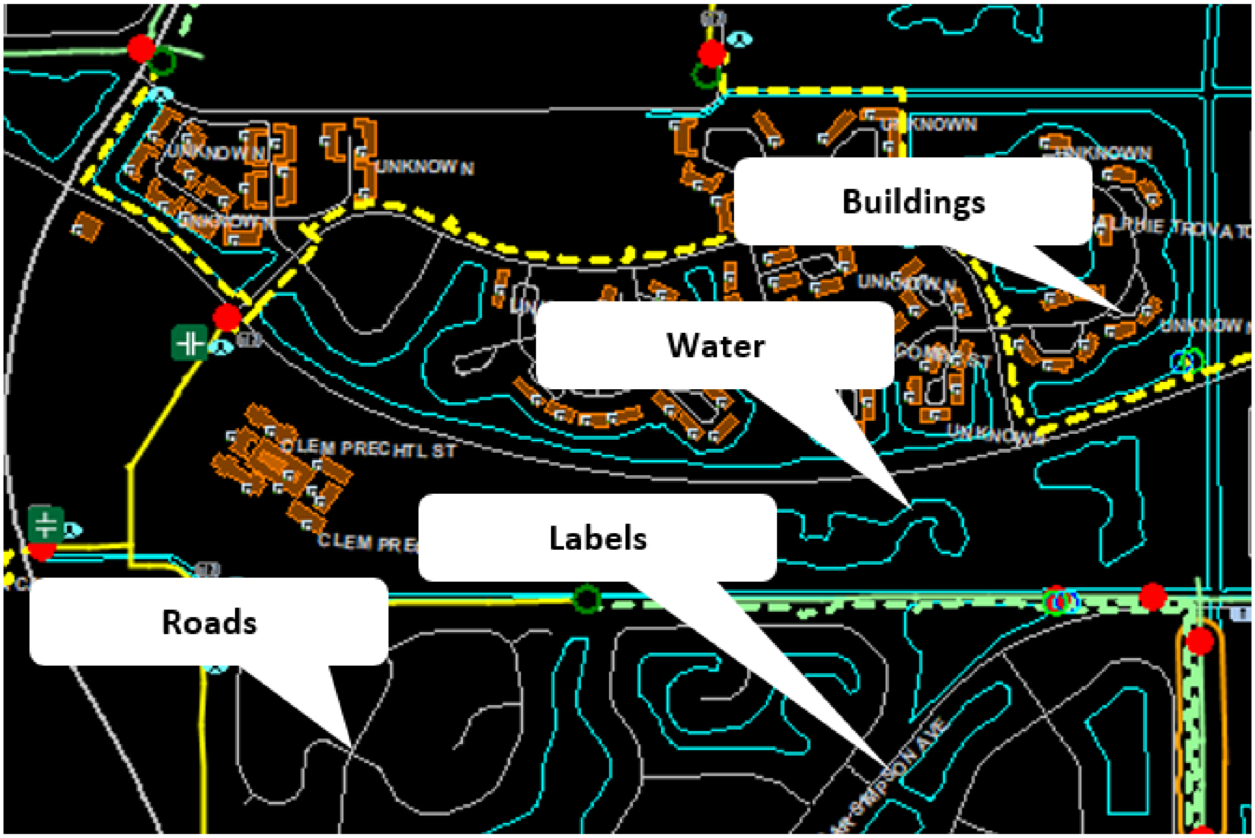


Figure 2. Background Shapes at Default Zoom Level

1.2. High Zoom Level View

This section provides a high-level overview of the electrical devices and halos that are displayed on the Geographic Network View at high zoom levels (i.e., zoom levels larger than 1). All devices such as distribution transformers become visible at high zoom levels, and the halos are applied to the individual elements. Minor background shapes also become visible at high zoom levels.

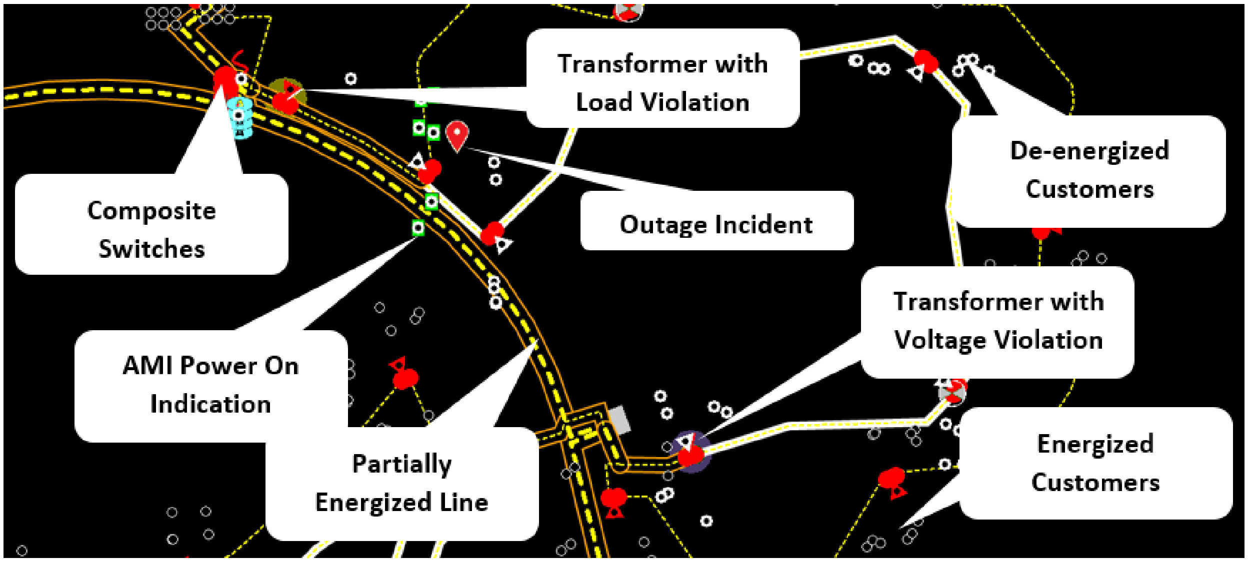


Figure 3. Network Elements at High Zoom Level

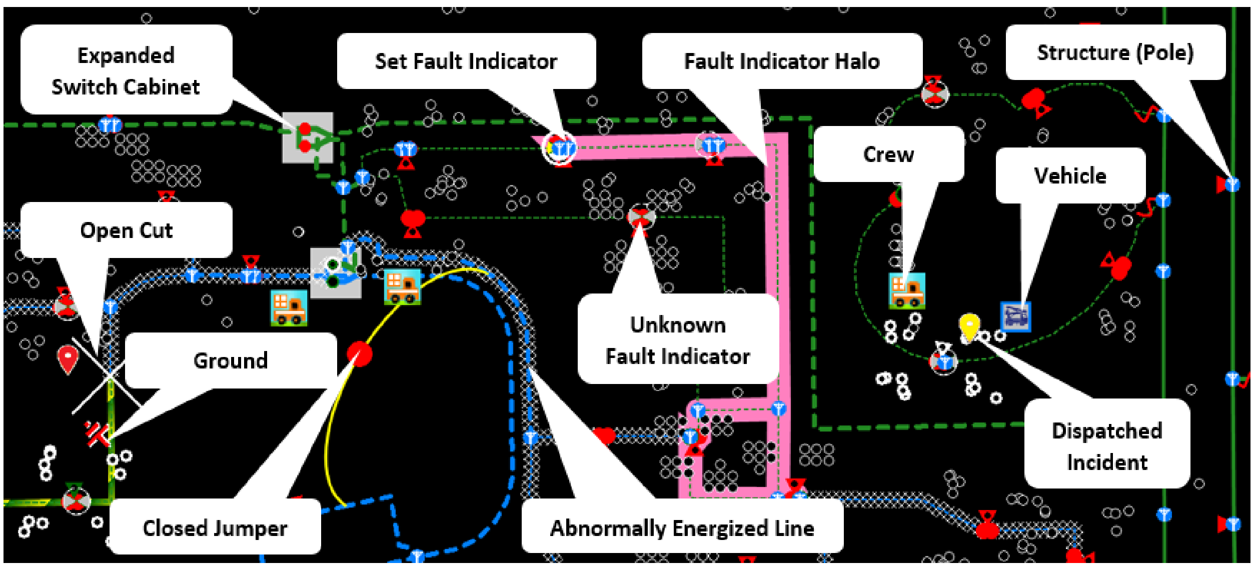


Figure 4. Network Elements at High Zoom Level

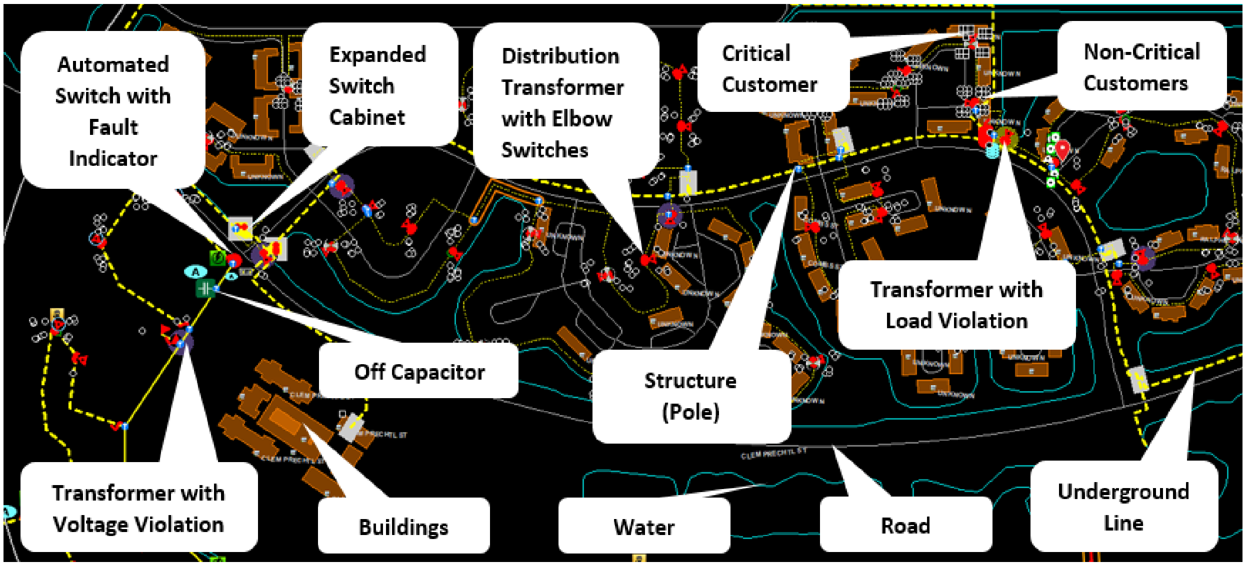


Figure 5. Network Elements at High Zoom Level

2. Electrical Devices

This chapter describes the vector symbols for the following electrical devices:

- Switch
- Transformer
- Substation
- UI Containers
- Capacitor
- Fault Indicator
- Energy Resource
- Other Devices

2.1. Switching Devices

This section presents vector graphics for switching devices:

- Switch
- Composite Switch
- Breaker
- Recloser
- Sectionalizer
- Fuse
- Earth/Ground Switch
- Elbow
- Switch State Decorations
- Switch Specific Open State Classifications

2.1.1. Switch

- Closed Switch
- Open Switch

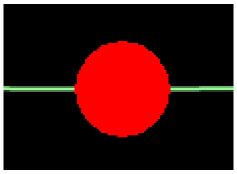
Configurable: Yes

ADMS Tool: Viewer Configuration Editor

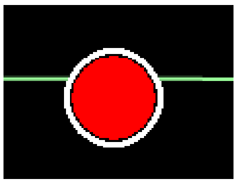
Reference Document: *Viewer Configuration Editor User Guide*

2.1.1.1. Closed Switch

Normally closed switch:

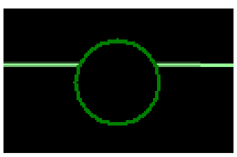


Abnormally closed switch:

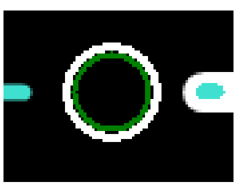


2.1.1.2. Open Switch

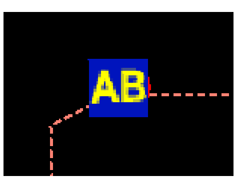
Normally open switch:



Abnormally open switch:



2.1.2. Composite Switch



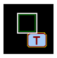
Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.1.3. Breaker

Closed Breaker



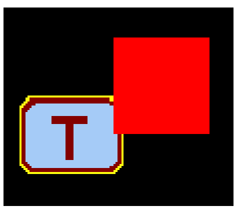
Abnormally Open Breaker

Configurable: Yes

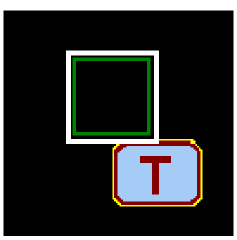
ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.1.3.1. Closed Breaker



2.1.3.2. Abnormally Open Breaker



2.1.4. Recloser

Open Recloser

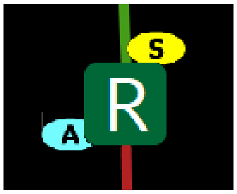
Closed Recloser

Configurable: Yes

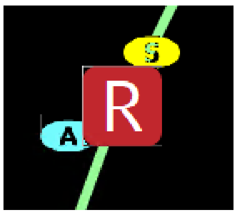
ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.1.4.1. Open Recloser



2.1.4.2. Closed Recloser



2.1.5. Sectionalizer

 Open Sectionalizer

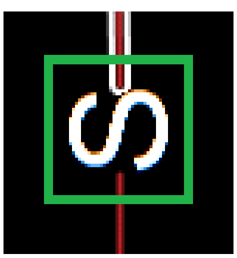
 Closed Sectionalizer

Configurable: Yes

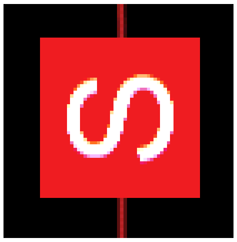
ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.1.5.1. Open Sectionalizer



2.1.5.2. Closed Sectionalizer



2.1.6. Fuse

Fuses are used to both protect and isolate the network. They behave similar to manually controlled switches, but show no indication of abnormal status. Fuse switches are always considered to have a "closed" normal state.

 Closed Fuse

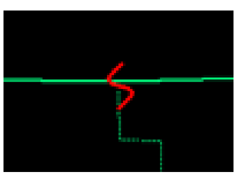
 Open Fuse

Configurable: Yes

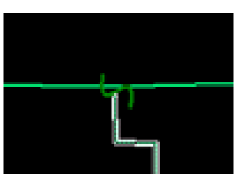
ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

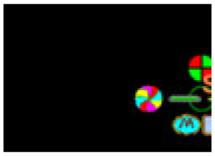
2.1.6.1. Closed Fuse



2.1.6.2. Open Fuse



2.1.7. Earth/Ground Switch

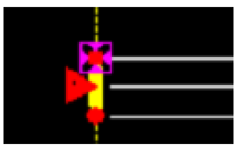


Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.1.8. Elbow



An elbow switch is located on both sides of the transformers in Underground Residential Distribution (URD) loops.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

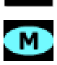
Reference Document: *Viewer Configuration Editor User Guide*

2.1.9. Switch State Decorations

SCADA Controllable


Pending Permanent State

Telemetered

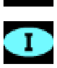


Manually Replaced

Not In Service


Abnormal Related

Bypassed



Isolated

Suspect Quality

Reclose Disabled

 Supervisory Cutoff On

 Block Close

 Ground Trip Disabled

 Loss of Voltage - Station

 Loss of Voltage - Feeder

 Dangerous Switching Condition

 Automated Scheme Team Enabled

 Automated Scheme Team Disabled

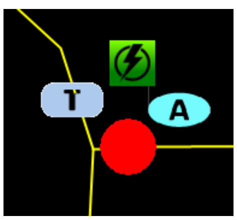
TEST Test Mode

Configurable: Yes (Decorations)

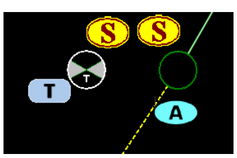
ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

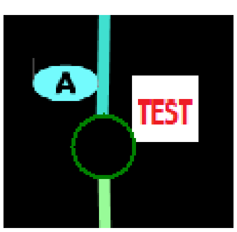
2.1.9.1. SCADA Controlled Switch in Normal State with Fault Indicator



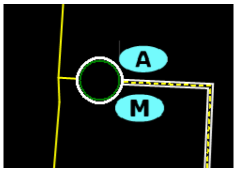
2.1.9.2. SCADA Controlled Switch in Suspect State with Fault Indicator



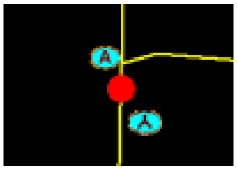
2.1.9.3. SCADA Controlled Switch in Test Mode



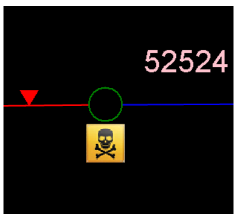
2.1.9.4. SCADA Controlled Switch in Manually Replaced State



2.1.9.5. SCADA Controlled Switch with Abnormal Status

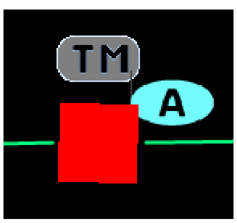


2.1.9.6. Dangerous Switching Condition



The Dangerous Switching Condition decoration appears on open switches/jumpers/cuts whose closing could result in a dangerous network condition, such as dynamic phase mismatch, phase angle mismatch, or nominal voltage mismatch. Note: Nominal voltage mismatch check is not applicable to cut.

2.1.9.7. Automated Scheme Team Member



The team enabled and team disabled decorations appear on switching devices (switches, breakers, reclosers, or sectionalizers) that are part of local peer-to-peer switching schemes used to automate the fault isolation and service restoration functions.

2.1.10. Switch Specific Open State Classifications

Abnormally open switch with the Temporary Open classification:

Abnormally open switch with the Bad Open classification:

Abnormally open switch with the Normal Open classification:

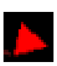
Configurable: Yes

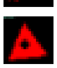
ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.2. Transformers

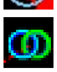
This section presents vector graphics for the following types of transformers:

 Distribution Transformer (Overhead)

 Distribution Transformer (Underground)

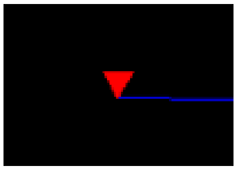
 Distribution Transformer with Embedded Generation

 Distribution Transformer with Connected Energy Resource

 Auto Transformer or Step Transformer

 Power Transformer (Station Transformer)

2.2.1. Distribution Transformer (Overhead)



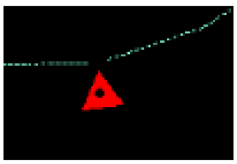
The transformer symbol orientation is adjusted for aesthetics and does not represent anything in the field.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.2.2. Distribution Transformer (Underground)



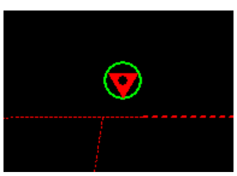
The black dot in the center of the symbol indicates that the distribution transformer is modeled as an underground device.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

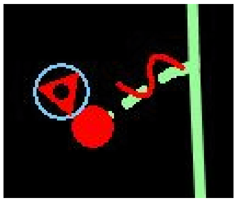
2.2.3. Distribution Transformer with Embedded Generation



A green circle around a distribution transformer indicates that loads with embedded generation (for example, homes with rooftop solar panels or energy storage) are connected to the transformer.

Configurable: No

2.2.4. Distribution Transformer with Connected Energy Resource

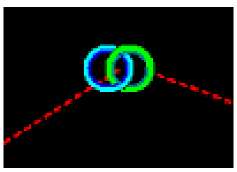


A white circle around a distribution transformer indicates that the distribution transformer has an energy resource connected to the secondary side.

Configurable: No

ADMS Tool: Viewer Configuration Editor

2.2.5. Auto Transformer or Step Transformer

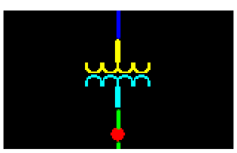


Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User guide*

2.2.6. Power Transformer (Station Transformer)



Configurable: Yes

ADMS Tool: Viewer Configuration Editor

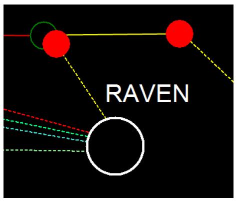
Reference Document: *Viewer Configuration Editor User Guide*

2.3. Substation

This section presents vector graphics for the following types of substation:

 Substation
 Mobile Substation

2.3.1. Substation

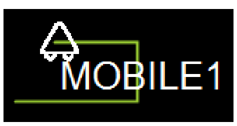


Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.3.2. Mobile Substation



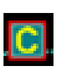
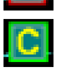
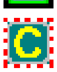
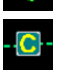
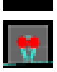
Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

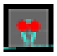
2.4. UI Containers

This section presents vector graphics for the following UI containers:

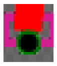
 Switch Cabinet in Connected State
 Switch Cabinet in Open State
 Switch Cabinet in Closed Abnormal State
 Switch Cabinet in Open Abnormal State
 Special Switch Cabinet
 Switch Cabinet Internals
 Vault in Connected State

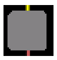
 Vault in Open State

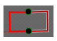
 Vault in Abnormal State

 Vault Internals

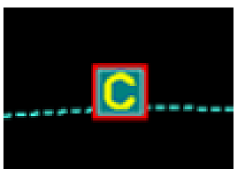
 Recloser UI Container

 Zoomed-In Recloser UI Container

 UI Container

 Zoomed-In UI Container

2.4.1. Switch Cabinet in Connected State



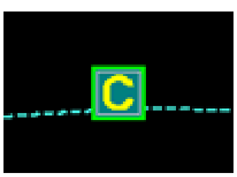
The red border around the switch cabinet indicates that all switches are closed.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.4.2. Switch Cabinet in Open State



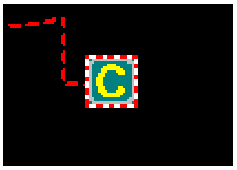
The green border around the switch cabinet indicates that there are open switches in the cabinet.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.4.3. Switch Cabinet in Closed Abnormal State



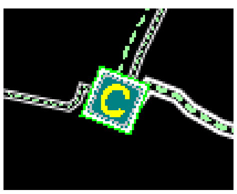
The red-white dashed border around the connected switch cabinet indicates that some of the switches are in the abnormally closed state.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.4.4. Switch Cabinet in Open Abnormal State



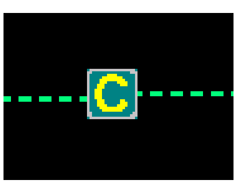
The green-white border around a disconnected switch cabinet indicates that some of the switches are in the abnormally open state.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.4.5. Special Switch Cabinet



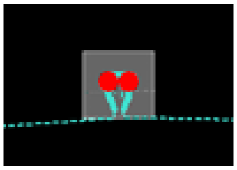
The white border around a switch cabinet indicates that this cabinet contains specific devices, such as transfer switches or pad mounted transformers.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.4.6. Switch Cabinet Internals



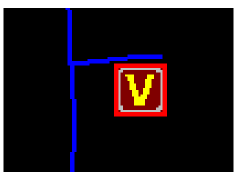
At a specified zoom level, the reconfigurable content of a switch cabinet appears instead of the cabinet symbol. Reconfigurable switches are those that can be used to tie feeders together. Therefore, fuses and switches that feed radial spurs are not considered to be “reconfigurable”.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.4.7. Vault in Connected State



Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

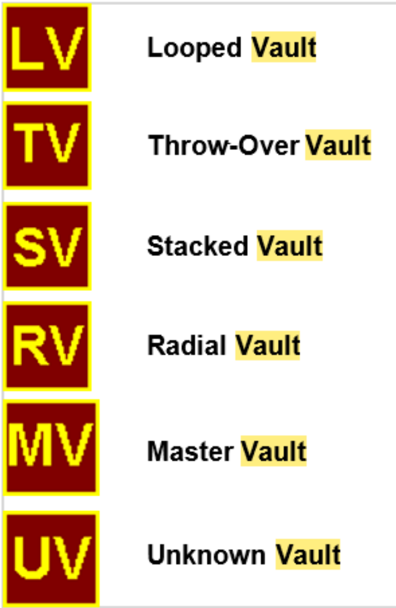
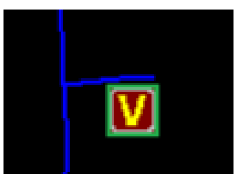


Figure 6. Vault Symbols

2.4.8. Vault in Open State

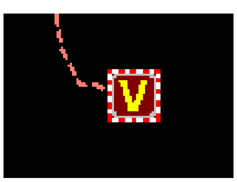


Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.4.9. Vault in Abnormal State

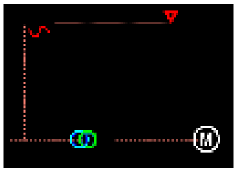


Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.4.10. Vault Internals

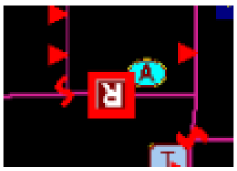


Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.4.11. Recloser UI Container



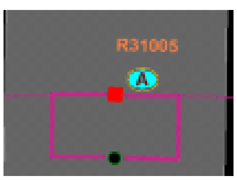
The recloser UI container usually contains a recloser with by-pass.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.4.12. Zoomed-In Recloser UI Container



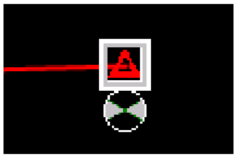
At a specified zoom level, the recloser's schematic view appears instead of the recloser UI container symbol.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.4.13. UI Container

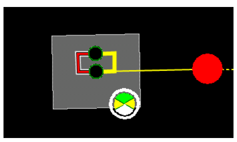


Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.4.14. Zoomed-In UI Container



At a specified zoom level, the UI container's schematic view appears instead of the UI container symbol.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.5. Capacitor

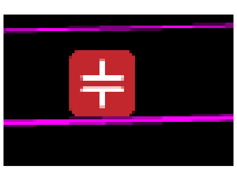
This section presents vector graphics for the following states of capacitor:

 Connected Capacitor

 Disconnected Capacitor

 Capacitor with Local Control

2.5.1. Connected Capacitor



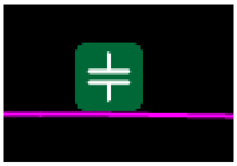
Capacitor in the ON position.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.5.2. Disconnected Capacitor



Capacitor in the OFF position.

Capacitor bank switching is performed manually in the field, and the dispatcher needs to update the status of the network operations model. Capacitor banks can be operated directly from the distribution Geographic Network View.

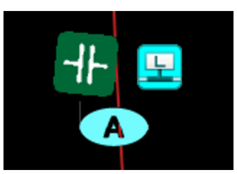
The capacitor bank symbol is selected by clicking on the short stub that is perpendicular to the main capacitor symbol. The system acknowledges the selection by showing the capacitor bank symbol in a magenta border.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.5.3. Capacitor with Local Control



The local control decoration  indicates that the capacitor is not controllable via SCADA.

Configurable: Yes (Decorations)

ADMS Tool: Viewer Configuration Editor

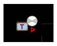
Reference Document: *Viewer Configuration Editor User Guide*

2.6. Fault Indicator

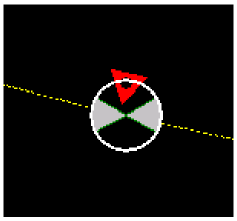
This section presents vector graphics for the following states of fault indicator:

 Manual Fault Indicator

 Telemetered Fault Indicator

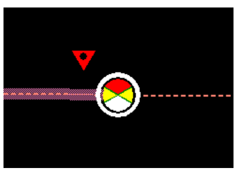
 Directional Fault Indicator

2.6.1. Manual Fault Indicator



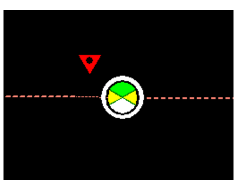
Transformer equipped with a manual fault indicator.

Set Fault Indicator



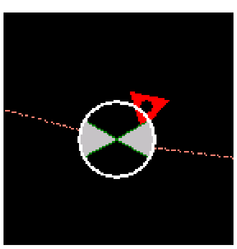
Fault indicator in the Set position, indicating a fault downstream.

Unset Fault Indicator



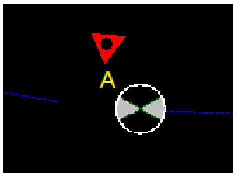
Fault indicator in the Unset position, indicating no fault conditions.

Unknown Fault Indicator



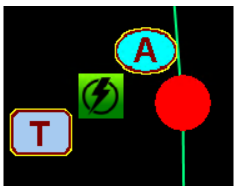
Fault indicator in the Unknown position. Note that only manual fault indicators can be in the Unknown state.

Phase-Agnostic Fault Indicator



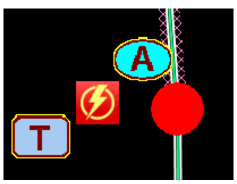
Phase-agnostic fault indicators do not detect the faulted phase. These fault indicators may appear with the “A” symbol above the device (configurable).

2.6.2. Telemetered Fault Indicator



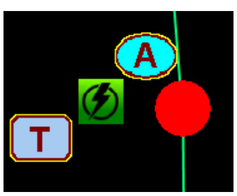
SCADA-controlled switch equipped with a telemetered fault indicator.

Set Telemetered Fault Indicator



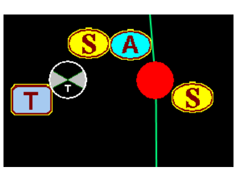
Fault indicator in the Set position, indicating a fault downstream.

Unset Telemetered Fault Indicator



Fault indicator in the Unset position, indicating no fault conditions.

Telemetered Fault Indicator in Suspect State



SCADA-controlled switch equipped with a telemetered fault indicator. The fault indicator and the switch are both in the Suspect state.

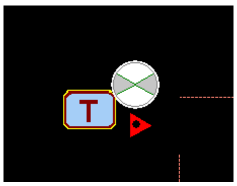
Fault indicators for super capacitor backup types may appear slightly different. If they are de-energized and suspect, the fault indicator icon shows its last known good state (Set or Unset), not the unknown state. The suspect decoration still appears.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

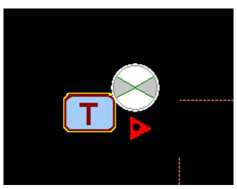
Reference Document: *Viewer Configuration Editor User Guide*

2.6.3. Directional Fault Indicator



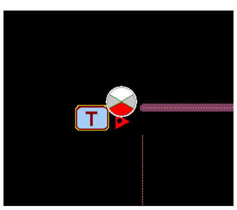
Transformer equipped with a telemetered directional fault indicator.

Directional Fault Indicator (Unset)



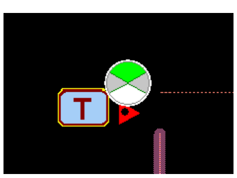
Directional fault indicator in the Unset position, indicating no fault conditions at both sides.

Directional Fault Indicator (Side 1)



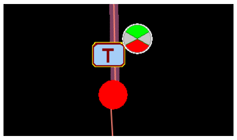
Directional fault indicator in the Side 1 (red) position, indicating a single-phase fault on Side 1 of the indicator.

Directional Fault Indicator (Side 2)



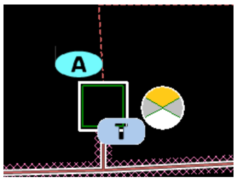
Directional fault indicator in the Side 2 (green) position, indicating a single-phase fault on Side 2 of the indicator.

Directional Fault Indicator (Both)



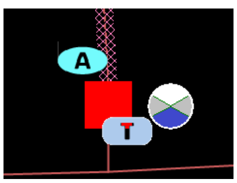
Directional fault indicator in both (Side 1 and Side 2) positions, indicating a multi-phase fault downstream from the fault indicator.

Directional Fault Indicator (Forward)



Directional fault indicator in the Forward (yellow) position, indicating a multi-phase fault downstream from the fault indicator.

Directional Fault Indicator (Reverse)



Directional fault indicator in the Reverse (blue) position, indicating a multi-phase fault upstream from the fault indicator.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.7. Energy Resource

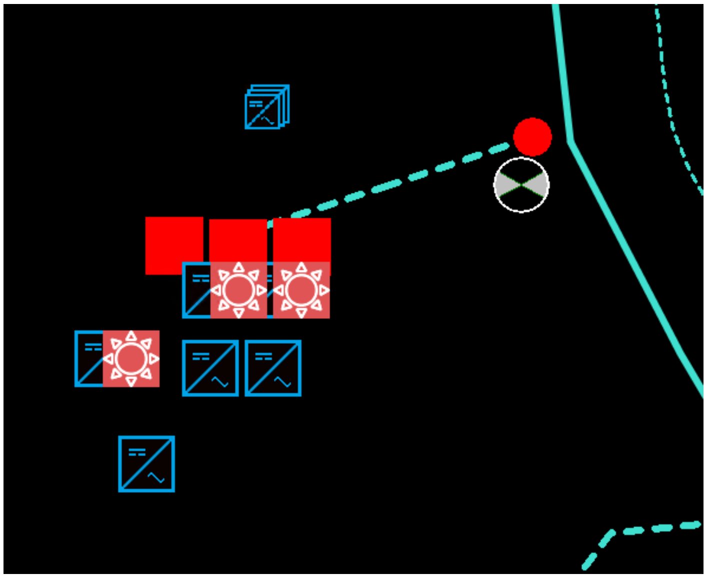
This section presents vector graphics for the following devices:

 Generator

 Distributed Energy Resource (DER)

 Inverter

 DER Storage Device Charge Level



2.7.1. Generator

Co-Generator:

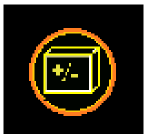
Wind:

Solar:

Micro Turbine:

Fuel Cell:

Battery Storage:



Not Set:

Configurable: Yes

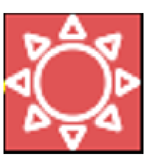
ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.7.2. Distributed Energy Resource (DER)

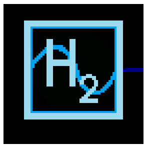
Wind:

Solar:

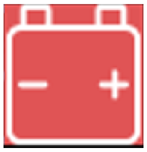


Micro Turbine:

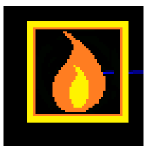
Fuel Cell:



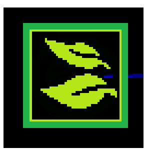
Battery Storage:



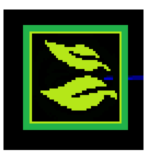
Gas:



Biomass:



Hydro:



EV Charging Station:

Others:

Cogeneration (specific bitmap is not assigned by default):

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.7.3. Inverter

Inverters and inverter groups define voltage, power factor, and deliverable and absorbable power characteristics of the associated Distributed Energy Resource (DER).

Configurable: Yes

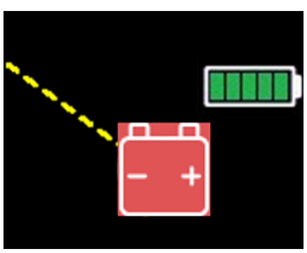
ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

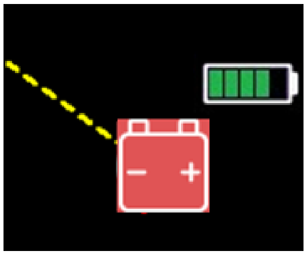
2.7.4. DER Storage Device Charge Level

DER battery storage device charge level is indicated by a decoration.

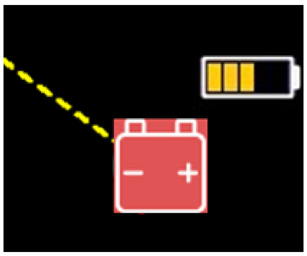
Full Charge:



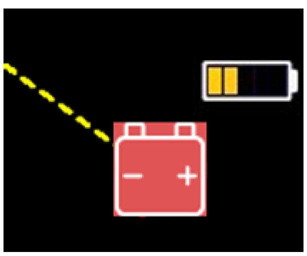
Battery Charged at 80%:



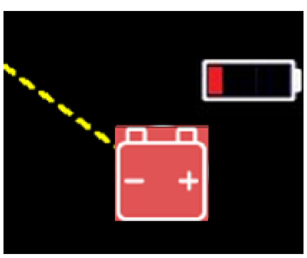
Battery Charged at 60%:



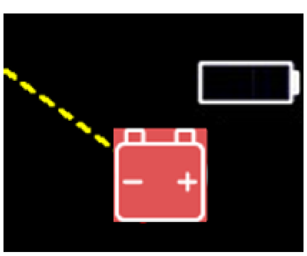
Battery Charged at 40%:



Battery Charged at 20%:



No Charge:



Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.8. Other Devices

This section presents vector graphics for the following devices:

 Reference Bus Interface

 Regulator

 Sensor Location

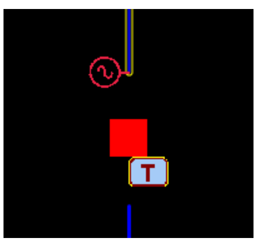
 Series Reactor

 Load (Primary Meter)

Network Protector

 Orphaned Device

2.8.1. Reference Bus Interface



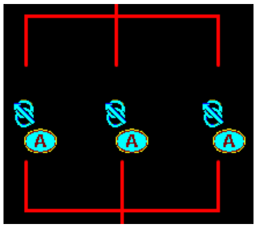
Reference bus interface objects specify the transmission voltage magnitude, voltage angle, and equivalent impedance parameters. The objects appear on the station internals view, upstream of station power transformers.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.8.2. Regulator



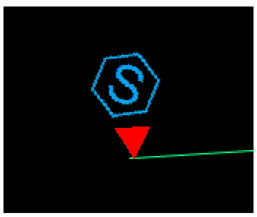
Regulators are used for feeder voltage regulation.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.8.3. Sensor Location



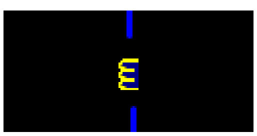
Sensor location is used for voltage and power flows monitoring. It sends measurements to SCADA for use with different DNAF applications.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.8.4. Series Reactor



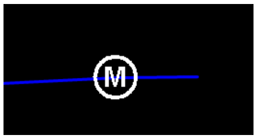
Series reactors are used to reduce short circuit currents in the distribution network.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.8.5. Load (Primary Meter)



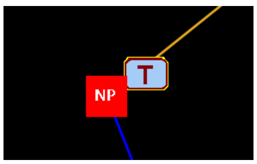
The load is used to indicate the major customer connection.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.8.6. Network Protector

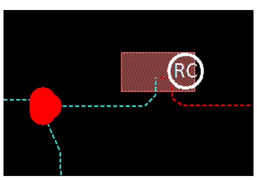


Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.8.7. Reserved Capacity

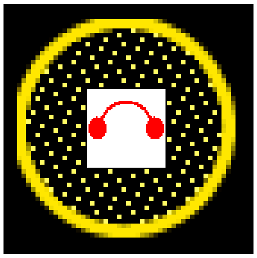


Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

2.8.8. Orphaned Device



Temporary network modifications that are not linked to any of the devices in production are called orphaned devices (for example, a cut on a line that was removed from the model). These devices are displayed on the Geographic Network View highlighted with a yellow dotted circle.

Configurable:

- **Halo:** No
- **Temporary Device Symbol:** Yes

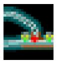
ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

3. Lines and Halos

This chapter describes the vector symbols for the following electrical devices and their states:

 Feeders

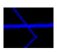
 Halos

3.1. Feeders

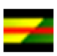
This section presents vector graphics for the following devices and their states:

 Overhead Feeder Fully Energized (Phase 3, 2, 1)

 Underground Feeder Fully Energized (Phase 3, 2, 1)

 Lateral

 De-energized Feeder

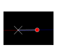
 Grounded Feeder

 Looped Feeder

 Parallel Looped Feeder

 Fed from Looped Feeder

 Traced Feeder

 Abnormally Energized Feeder

 Partially Energized Feeder

 Feeder with Opposing Flow

 Future

 Predicted Out Feeder

 Feeder Voltage Colors

3.1.1. Overhead Feeder Fully Energized (Phase 3, 2, 1)

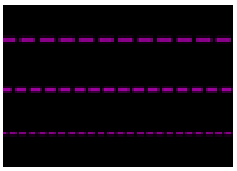
Feeders are different colors to distinguish them from their neighbors. Colors and styles are configured in the Viewer Configuration Editor.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

3.1.2. Underground Feeder Fully Energized (Phase 3, 2, 1)



Undeground Loop Color – Phase A

Undeground Loop Color – Phase B

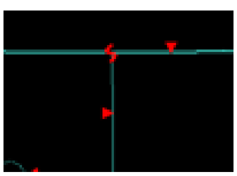
Undeground Loop Color – Phase C

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

3.1.3. Lateral



Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

3.1.4. De-energized Feeder

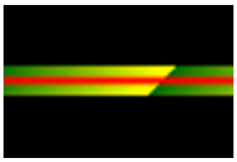
The white halo indicates that all the line phases are de-energized or in the abnormal state. The color and style of each halo state are configured in the Viewer Configuration Editor.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

3.1.5. Grounded Feeder



The feeder with applied temporary ground is shown with green halo, or with green halo with yellow stripe, depending on the configuration settings.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

3.1.6. Looped Feeder

When a single feeder contains a network loop, this halo is used to highlight the main loop.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

3.1.7. Parallel Looped Feeder

When the feeder is energized from two or more sources, this halo is used to highlight the network part between the energization points.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

3.1.8. Fed from Looped Feeder

When a feeder is parallel looped with another feeder, this halo is used to highlight the network part that is not part of the main energization loop, but is connected to it.

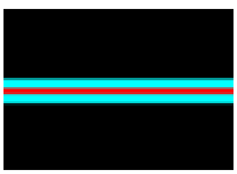
Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

3.1.9. Traced Feeder

The traced feeder is highlighted with cyan halo.

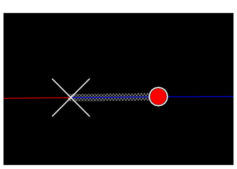


Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

3.1.10. Abnormally Energized Feeder



An abnormally energized feeder is presented with its line color changed to that of the energizing circuit with white cross-hatching.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

3.1.11. Partially Energized Feeder

When a feeder or line is not fully energized (there are de-energized phases), it is highlighted with orange halo.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

3.1.12. Feeder with Opposing Flow

This halo indicates a directional flow mismatch between some of the phases in this line.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

3.1.13. Future Equipment

Future equipment (not yet commissioned) is shown with an orange halo.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

3.1.14. Future Remove Equipment

Future Remove equipment (not yet decommissioned) is shown with an yellow halo.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

3.1.15. Removed Equipment

Removed equipment that has been decommissioned is shown with an brown halo.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

3.1.16. Predicted Out Feeder

Predicted out feeders are shown with an orange halo.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

3.1.17. Feeder Voltage Colors

230 KV

138 KV

23 KV

13 KV

132 KV

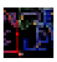
7 KV

79 KV

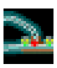
0 KV

3.2. Halos

This section presents vector graphics for the following states of the electrical devices:

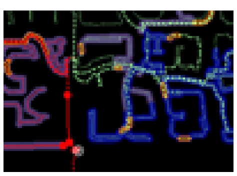
 Voltage Violation Halos

 Overload Halos

 Fault Halos

 Capacity Margin Halo

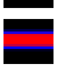
3.2.1. Voltage Violation Halos



Dark blue indicates a violation of the maximum voltage limit. Purple indicates a violation of the minimum voltage limit.

At normal zoom levels (whole feeder view), the violation halo is applied to the entire switchable section that contains the device that is in violation. At high zoom levels, where all the devices such as distribution transformers are visible, the halos are applied to the individual elements.

 Low Voltage

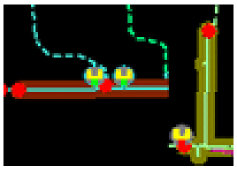
 High Voltage

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

3.2.2. Overload Halos



A yellow halo indicates a level 1 overload. A dark yellow halo indicates a level 2 overload. An orange halo indicates a level 3 overload.

At normal zoom levels (whole feeder view), the overload halo is applied to the entire switchable section that contains the device that is in violation. At high zoom levels, where all the devices such as distribution transformers are visible, the halos are applied to the individual elements.

 Overload Violation Level 1 (High)

 Overload Violation Level 2 (Very High)

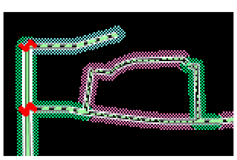
 Overload Violation Level 3 (Extreme)

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

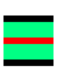
Reference Document: *Viewer Configuration Editor User Guide*

3.2.3. Fault Halos

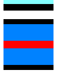


The potential fault location halo is defined in three colors, indicating the levels of likelihood that the fault is located on that segment of the feeder. The green halo (level 1) indicates the most likely fault location.

The fault indicator halo highlights the segments of a feeder affected by a fault based on the fault indicator measurements.

 Potential Fault Level 1

 Potential Fault Level 2

 Potential Fault Level 3

 Fault Indicator

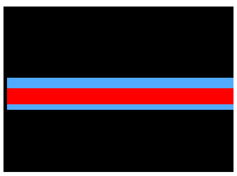
Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

3.2.4. Capacity Margin Halo

This halo shows capacity margin violations.



Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

4. Temporary Modification Symbols

This chapter presents vector graphics for the following temporary modification symbols:

 Jumper

 Temporary Ground

 Fault

 Cut

 Temporary Switch

 Elbow Switch

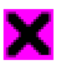
 Temporary Recloser

 Temporary Sectionalizer

Gang Switch

 Disconnect

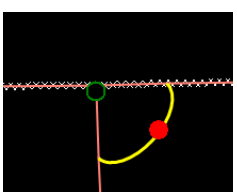
 Fuse

 Temporary Marks

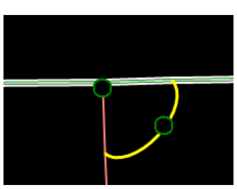
4.1. Jumper

Jumpers are temporary switch connections between parts of feeder backbones or laterals. They are used to energize laterals or portions of feeders that have been isolated because of faults or maintenance.

Closed Jumper:



Open Jumper:

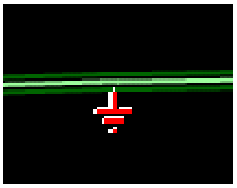


Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

4.2. Temporary Ground



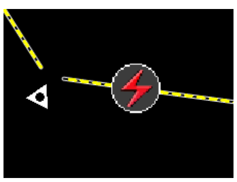
Temporary grounds can only be placed on the de-energized sections of a line.

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

4.3. Fault



 Fault Placed by a User

 Fault Placed by a User, Solved

 Predicted Fault

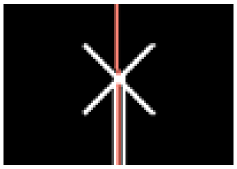
 Predicted Fault, Solved

Configurable: No

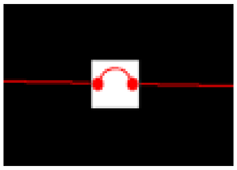
4.4. Cut

A cut is a temporary break of one or more phases. It can be placed on any type of line.

Open Cut:



Closed Cut:



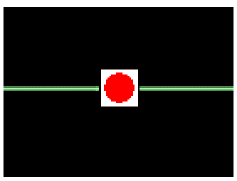
Configurable: Yes

ADMS Tool: Viewer Configuration Editor

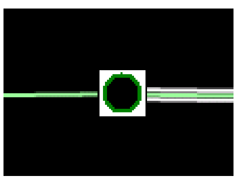
Reference Document: *Viewer Configuration Editor User Guide*

4.5. Temporary Switch

Closed Temporary Switch:



Open Temporary Switch:



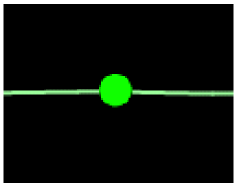
Configurable: Yes

ADMS Tool: Viewer Configuration Editor

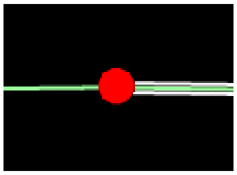
Reference Document: *Viewer Configuration Editor User Guide*

4.6. Elbow Switch

Closed Elbow Switch:



Open Elbow Switch:



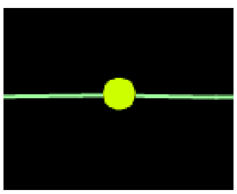
Configurable: Yes

ADMS Tool: Viewer Configuration Editor

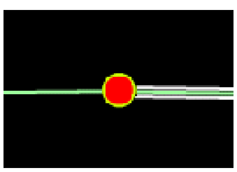
Reference Document: *Viewer Configuration Editor User Guide*

4.7. Temporary Recloser

Closed Recloser:



Open Recloser:



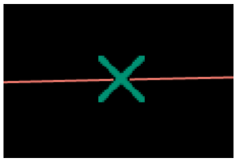
Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

4.8. Temporary Sectionalizer

Closed Sectionalizer:



Open Sectionalizer:

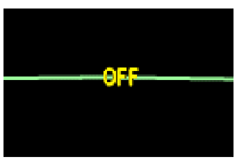
Configurable: Yes

ADMS Tool: Viewer Configuration Editor

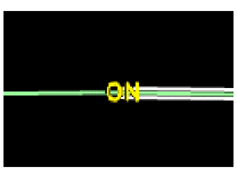
Reference Document: *Viewer Configuration Editor User Guide*

4.9. Gang Switch

Closed Gang Switch:



Open Gang Switch:



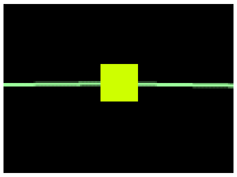
Configurable: Yes

ADMS Tool: Viewer Configuration Editor

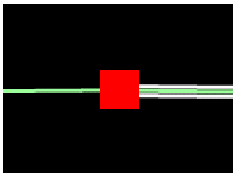
Reference Document: *Viewer Configuration Editor User Guide*

4.10. Disconnect

Closed Disconnect:



Open Disconnect:



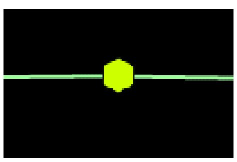
Configurable: Yes

ADMS Tool: Viewer Configuration Editor

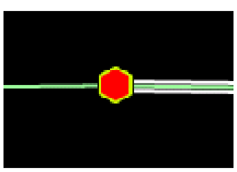
Reference Document: *Viewer Configuration Editor User Guide*

4.11. Fuse

Closed Fuse:



Open Fuse:



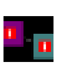
Configurable: Yes

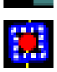
ADMS Tool: Viewer Configuration Editor

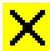
Reference Document: *Viewer Configuration Editor User Guide*

4.12. Temporary Marks

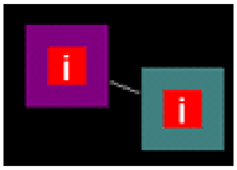
This section describes the following temporary marks:

 Move Mode

 Mark

 Select

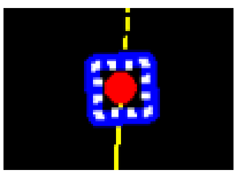
4.12.1. Move Mode



Temporary objects on the Geographic Network View can be moved using the pre-defined function keys combination. In this picture, an incident symbol is in the move mode (the incident location is changing manually). Function keys combinations are configured in the Viewer Configuration Editor.

Configurable: No

4.12.2. Mark



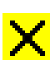
Some devices and temporary objects on the Geographic Network View can be marked using the toolbar or right-click menu option. This picture illustrates the marked switch.

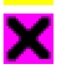
Configurable: One of three mark symbols can be used.

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

4.12.3. Select

 Previously Selected Object

 Currently Selected Object

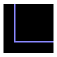
Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

5. Background Layers

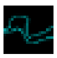
This chapter presents vector graphics for the following background layers of the Geographic Network View:

 Boundaries

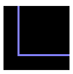
 Building Footprints

 Roads

 Roads with Labels

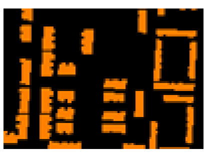
 Water

5.1. Boundaries



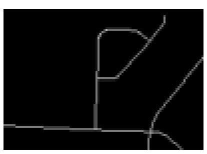
Configurable: No

5.2. Building Footprints



Configurable: No

5.3. Roads



The major and minor roads are configured to appear at various zoom levels, to manage clutter.

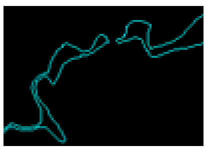
Configurable: No

5.4. Roads with Labels

Road labels are configured to appear at a certain zoom level, to manage clutter.

Configurable: No

5.5. Water



Only the water boundaries are shown.

Configurable: No

6. Dynamic Objects

This chapter describes the following dynamic objects that appear on the Geographic Network View when certain conditions occur (outages, customer calls, etc.).

 Incidents

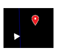
 Customers

 Crews

 Vehicles

 Annotations

6.1. Incidents



 Planned Outage

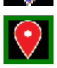
 Unassigned Incident

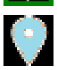
 Assigned Incident

 Evaluating Incident

 Evaluated Incident

 Pending Clear Incident

 Predicted Stable Incident

 Non-Outage Incident

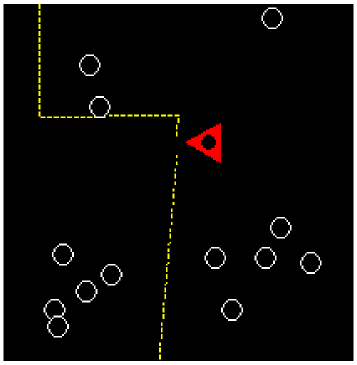
 Unassociated Call

Configurable: Yes

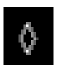
ADMS Tool: Coded Translation Editor

Reference Document: *Coded Translation Editor User Guide*

6.2. Customers

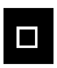


Customers are configured to appear at a certain zoom level, to manage clutter.

 Critical Customer

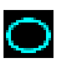
 Normal Customer

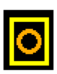
 Inactive Customer

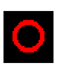
 Not Connected Customer

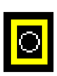
 De-energized Customer

 Hazardous Customer

 Traced Customer

 Predicted Out Customers

 Suspended Customer

 Customer Call

 Automated Meter (AMI) Ping Report

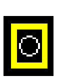
 Automated Meter (AMI) State

Configurable: Yes

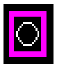
ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

6.2.1. Customer Call

 Single customer non-priority call, not confirmed outage

 Single customer call, confirmed outage

 Single customer priority call, not confirmed outage

 Single customer priority call, confirmed outage

6.2.2. Automated Meter (AMI) Ping Report

 AMI Ping Sent

 AMI Ping In

 AMI Ping Out

 AMI Ping Error

6.2.3. Automated Meter (AMI) State

 AMI In

 AMI Out

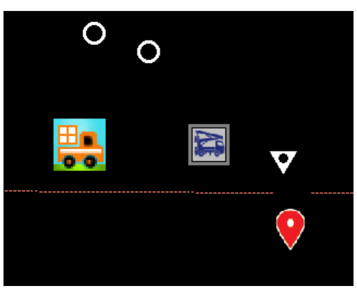
 AMI Operation Error

 Low Meter Voltage

 High Meter Voltage

 Nominal Meter Voltage

6.3. Crews



 First Responder Crew

 Foreman

 Evaluator

 Local Lineman

 Overhead Crew

 Serviceman

 Underground Crew

Configurable: Yes

ADMS Tool: Coded Translation Editor

Reference Document: *Coded Translation Editor User Guide*

6.4. Vehicles

 Pickup

 Bucket Truck

 Derrick

 Digger

 Boom Truck

 Service Truck

 Survey Truck

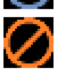
Configurable: Yes

ADMS Tool: Coded Translation Editor

Reference Document: *Coded Translation Editor User Guide*

6.5. Damage Assessment Locations

 Repaired Incident

 Not Repaired Incident

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

6.6. Annotations

Key Customer

Trouble Ticket

Operation Note

Switching

Data Quality

Trouble

Crew

Follow Up

Outage

Call

Lightning

Safety

Fixed

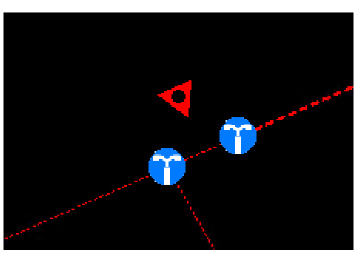
Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*

6.7. Structures

Structures are non-electrical objects, such as poles, street lights, or restricted area boundaries. Structure objects are exported from GIS.



Street Light

Pole

Call Box

Configurable: Yes

ADMS Tool: Viewer Configuration Editor

Reference Document: *Viewer Configuration Editor User Guide*